

PRESSURE VESSEL TECHNICAL DOCUMENTATION

IDENTITY INFORMATION

Company_Name: Meridian Offshore Energy Solutions LLC

Field_Engineer: James R. Hutchinson

Field_Engineer_Id: MOS-ENG-4782

Document_Version: 2.3.1

Machine_Name: Mud Gas Separator System - Platform Delta-7

Machine_Id: MGS-D7-2019-001

Installation_Date: 2019-03-15

Number_Of_Units_In_Machine: 3

UNIT 1: PRIMARY SEPARATION VESSEL

Unit_Name: Primary Mud Gas Separator Vessel

Unit_Id: PMSV-D7-001

Manufacture_Date: 2018-11-22

Manufacturer_Company_Name: Weatherford International Solutions

Manufacturer_Company_Id: WIS-MFG-7741

Inventory_Quantity: 2 units in stock

Supplier_Information: Weatherford Houston Distribution Center, Lead Time: 6-8 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 12,000 feet TVD

- **Operating_Pressure_PSI:** 500 PSI (rated), 425 PSI (normal operating)

- **Drilling_Fluid_Compatibility:** Oil-based mud (OBM), synthetic-based mud (SBM), water-based mud (WBM)

- **Known_Error_Codes:** ERR-401 (pressure relief valve malfunction), ERR-203 (internal baffle corrosion)

- **Replacement_Part_Number:** WIS-PMSV-500-12K-Rev3

- **Maintenance_Schedule:** Quarterly inspection, semi-annual hydrostatic testing, annual internal cleaning

SECTION 1 - USE CASE

- Separates entrained gas from drilling mud returns during well operations

- Reduces atmospheric emissions and recovers hydrocarbon gas for flare or sales line

- Maintains mud weight stability by removing gas cut mud before return to active system
- Operates continuously during drilling, tripping, and circulation operations
- Critical safety barrier preventing gas-cut mud from reaching mud tanks

SECTION 2 - FAILURE CAUSES

- **ERR-401 (35% frequency):** Pressure relief valve spring fatigue from cyclic loading; manifests as inability to maintain setpoint
- **ERR-203 (28% frequency):** Internal baffle corrosion from H₂S exposure in sour gas environments; reduces separation efficiency
- **ERR-512 (18% frequency):** Inlet nozzle erosion from high-velocity mud jets; causes bypass of unseparated gas
- **ERR-307 (12% frequency):** Outlet ball valve stiction from drilling fluid residue accumulation
- **ERR-615 (7% frequency):** Structural fatigue cracks in vessel shell from pressure cycling and vibration

SECTION 3 - SOLUTIONS

- **ERR-401 Mitigation:** Replace pressure relief valve cartridge per API RP 54 Section 8.2.1; perform bench testing before reinstallation; inspect spring for permanent set
- **ERR-203 Mitigation:** Apply internal epoxy coating during scheduled maintenance; increase inspection frequency in H₂S service; consider duplex stainless steel baffle upgrade
- **ERR-512 Mitigation:** Install inlet flow diffuser to reduce mud jet velocity per OSHA 1910.119(l); inspect nozzle bore quarterly; replace if erosion exceeds 1/8 inch
- **ERR-307 Mitigation:** Perform quarterly ball valve actuation cycles; apply penetrating oil if stiction detected; replace valve if movement cannot be restored
- **ERR-615 Mitigation:** Conduct annual ultrasonic thickness survey per ASME Section VIII; perform dye penetrant inspection of high-stress areas; retire vessel if wall thickness drops below 0.375 inches

SECTION 4 - INVENTORY DATA

- **Current Stock:** 2 complete units available at Houston facility; 8 replacement relief valve cartridges in stock
- **Reorder Threshold:** Maintain minimum 1 complete unit; order replacement when stock reaches 1 unit
- **Supplier:** Weatherford International Solutions, Houston TX; secondary supplier: Smith Services Equipment, Galveston TX (10-week lead time)
- **Lead Time & Cost:** Primary supplier 6-8 weeks, \$47,500 per unit; expedited shipping available at 15% premium

UNIT 2: SECONDARY DEGASSING VESSEL

Unit_Name: Secondary Degassing Separator Vessel

Unit_Id: SDSV-D7-002

Manufacture_Date: 2018-12-10

****Manufacturer_Company_Name:** Weatherford International Solutions**

****Manufacturer_Company_Id:** WIS-MFG-7741**

****Inventory_Quantity:** 1 unit in stock**

****Supplier_Information:** Weatherford Houston Distribution Center, Lead Time: 6-8 weeks**

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:** 12,000 feet TVD**
- ****Operating_Pressure_PSI:** 150 PSI (rated), 100 PSI (normal operating)**
- ****Drilling_Fluid_Compatibility:** Oil-based mud (OBM), synthetic-based mud (SBM), water-based mud (WBM)**
- ****Known_Error_Codes:** ERR-204 (weir plate separation), ERR-418 (outlet strainer blockage)**
- ****Replacement_Part_Number:** WIS-SDSV-150-12K-Rev2**
- ****Maintenance_Schedule:** Monthly strainer cleaning, quarterly weir plate inspection, semi-annual vessel internal inspection**

SECTION 1 - USE CASE

- Provides secondary gas separation for mud returning from primary separator
- Reduces residual gas content to <2% by volume before mud enters reserve pit or treatment system
- Operates at lower pressure than primary vessel to maximize gas release
- Handles mud overflow during surge conditions and equipment shutdowns
- Protects downstream mud treatment equipment from gas interference

SECTION 2 - FAILURE CAUSES

- ****ERR-204 (42% frequency):** Weir plate separation from vibration and thermal cycling; causes mud bypass and loss of separation efficiency**
- ****ERR-418 (31% frequency):** Outlet strainer blockage from fine solids accumulation; increases backpressure and reduces throughput**
- ****ERR-506 (15% frequency):** Vent line blockage from mud carryover; prevents gas escape and causes pressure buildup**
- ****ERR-309 (8% frequency):** Corrosion of internal support structure from prolonged mud contact**
- ****ERR-711 (4% frequency):** Gasket degradation in outlet flange from thermal stress**

SECTION 3 - SOLUTIONS

- ****ERR-204 Mitigation:** Inspect weir plate welds quarterly per ASME Section VIII; re-weld if separation detected; consider welded baffle upgrade to eliminate bolted design**
- ****ERR-418 Mitigation:** Clean outlet strainer daily during operations; replace strainer element if differential pressure exceeds 5 PSI; upgrade to 200-micron element if solids loading is high**
- ****ERR-506 Mitigation:** Install vent line mud trap per API RP 54 Section 7.3; inspect trap daily; flush vent line with compressed air if blockage suspected**
- ****ERR-309 Mitigation:** Apply internal epoxy coating during maintenance; increase inspection frequency; consider stainless steel support structure upgrade**

- **ERR-711 Mitigation:** Replace outlet flange gasket during each maintenance interval; use PTFE-lined gaskets rated for 150 PSI service; torque bolts to 85 ft-lbs per API 6A

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete unit available; 12 replacement strainer elements in stock; 6 weir plate assemblies available

- **Reorder Threshold:** Order replacement unit when stock reaches zero; maintain 12-unit strainer element inventory

- **Supplier:** Weatherford International Solutions, Houston TX; alternative: Halliburton Pressure Control, Tomball TX (8-week lead time)

- **Lead Time & Cost:** Primary supplier 6-8 weeks, \$32,800 per unit; strainer elements \$185 each; weir plate assemblies \$2,400 each

UNIT 3: EMERGENCY DUMP VESSEL

Unit_Name: Emergency Dump Separator Vessel

Unit_Id: EDSV-D7-003

Manufacture_Date: 2019-01-08

Manufacturer_Company_Name: Weatherford International Solutions

Manufacturer_Company_Id: WIS-MFG-7741

Inventory_Quantity: 1 unit in stock

Supplier_Information: Weatherford Houston Distribution Center, Lead Time: 6-8 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 12,000 feet TVD

- **Operating_Pressure_PSI:** 50 PSI (rated), 25 PSI (normal operating)

- **Drilling_Fluid_Compatibility:** Oil-based mud (OBM), synthetic-based mud (SBM), water-based mud (WBM)

- **Known_Error_Codes:** ERR-105 (dump valve stiction), ERR-602 (float chamber malfunction)

- **Replacement_Part_Number:** WIS-EDSV-50-12K-Rev1

- **Maintenance_Schedule:** Weekly dump valve actuation test, monthly float chamber inspection, quarterly full system test

SECTION 1 - USE CASE

- Provides emergency mud dump capability during well control events or equipment failures

- Automatically diverts mud to reserve pit when pressure exceeds setpoint via float-operated dump valve

- Prevents overpressurization of mud system and protects downstream equipment from damage

- Operates in standby mode during normal drilling; activates only during emergency conditions

- Critical safety system per OSHA 1910.119 Process Safety Management requirements

SECTION 2 - FAILURE CAUSES

- **ERR-105 (48% frequency):** Dump valve stiction from drilling fluid residue and mineral deposits; prevents automatic dump activation during emergency
- **ERR-602 (32% frequency):** Float chamber malfunction from corrosion or mechanical damage; causes loss of pressure sensing capability
- **ERR-208 (12% frequency):** Dump line blockage from mud solids accumulation; restricts emergency mud flow
- **ERR-410 (5% frequency):** Pressure gauge failure; prevents operator awareness of system status
- **ERR-801 (3% frequency):** Structural corrosion of vessel exterior from saltwater spray exposure

SECTION 3 - SOLUTIONS

- **ERR-105 Mitigation:** Perform weekly manual dump valve actuation per OSHA 1910.119(j); apply penetrating oil if stiction detected; replace valve cartridge if movement cannot be restored within 24 hours
- **ERR-602 Mitigation:** Inspect float chamber monthly for corrosion and mechanical damage; replace float assembly if buoyancy is compromised; perform hydrostatic test after replacement
- **ERR-208 Mitigation:** Flush dump line with compressed air daily; install dump line strainer with 500-micron element; clean strainer weekly during operations
- **ERR-410 Mitigation:** Calibrate pressure gauge quarterly per API RP 54 Section 6.1; replace gauge if accuracy exceeds $\pm 2\%$ of full scale; install redundant gauge for critical monitoring
- **ERR-801 Mitigation:** Apply marine-grade epoxy coating to exterior surfaces annually; inspect for coating degradation quarterly; replace corroded sections if wall thickness loss exceeds 0.25 inches

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete unit available; 4 dump valve cartridges in stock; 6 float chamber assemblies available
- **Reorder Threshold:** Order replacement unit when stock reaches zero; maintain minimum 4 dump valve cartridges and 6 float assemblies
- **Supplier:** Weatherford International Solutions, Houston TX; backup supplier: Superior Energy Services, New Orleans LA (9-week lead time)
- **Lead Time & Cost:** Primary supplier 6-8 weeks, \$28,500 per unit; dump valve cartridges \$1,850 each; float chamber assemblies \$3,200 each

DOCUMENT CERTIFICATION

Prepared By: James R. Hutchinson, Field Engineer

Reviewed By: Margaret T. Chen, Senior Technical Manager (MOS-MGR-2156)

Approval Date: 2024-01-18

Next Review Date: 2025-01-18

Compliance Status: ✓ OSHA 1910 Compliant | ✓ API RP 54 Compliant | ✓ ASME Section VIII Compliant

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